

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\
a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-ae948d57e44516647.jsonl`
- SHA-256: `449ad3ee51f337516f6aa9f2e226d5e4403fea8421764ba83f27444287c6b9a7`
- Source modified: `2026-05-31T04:22:15+00:00`
- Imported at: `2026-07-05T16:48:21+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:06.025Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"olmOCR 2 is built on Qwen2.5-VL-7B-Instruct, a 7B-parameter vision-language model, fine-tuned then RL-trained; this anchors its VRAM/hardware footprint to the 7B Qwen2.5-VL tier (roughly 16-24 GB GPU class at FP16, less when quantized)."

****Source:**** <https://arxiv.org/html/2510.19817v1> (primary)

****Supporting quote:**** "We start with a Qwen2.5-VL-7B-Instruct model that has been fine-tuned on olmOCR-mix-1025."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:10.016Z)

I'll verify this claim about olmOCR 2's architecture and VRAM footprint. Let me start by loading the necessary tools and examining the primary source.

3. assistant (2026-05-31T04:21:13.680Z)

Now let me fetch the primary source to verify the supporting quote and the architectural claims.

4. user (2026-05-31T04:21:21.144Z)

olmOCR 2 Base Model and Training Details

****Base Model:****

olmOCR 2 is built on Qwen2.5-VL-7B-Instruct. The paper states: "We start with a Qwen2.5-VL-7B-Instruct model that has been fine-tuned on olmOCR-mix-1025."

****Supervised Fine-Tuning:****

The model undergoes initial fine-tuning on olmOCR-mix-1025, a dataset containing 267,962 pages from over 100,000 PDFs with improved formatting and equation handling.

****Reinforcement Learning Training:****

The paper describes the RL approach: "We apply Group Relative Policy Optimization (GRPO) to olmOCR using our synthetic verifiable unit tests as binary-valued reward signals." Training occurs on olmOCR2-synthmix-1025 (2,186 PDF pages with 30,381 test cases) for one epoch using "an 8xH100 GPU node," with 28 completions generated per document.

****Model Ensembling:****

Rather than a single trained model, the final olmOCR 2 uses weight averaging: "we train six models with different random seeds, and soup their weights at the end," combining both token-level and sequence-level importance sampling approaches.

****Hardware Requirements:****

Training utilized 8 H100 GPUs, though no VRAM specifications or inference requirements are explicitly stated.

5. user (2026-05-31T04:21:24.942Z)

Web search results for query: "olmOCR 2 Qwen2.5-VL-7B base model reinforcement learning olmOCR-mix-1025"

Links: [{"title": "[Literature Review] olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://www.themoonlight.io/en/review/olmocr-2-unit-test-rewards-for-document-ocr"}, {"title": "olmOCR 2 7B - Open Source OCR Model Guide 2025 - Apatero Blog | Apatero Blog - Open Source AI & Programming Tutorials", "url": "https://apatero.com/blog/olmocr-2-7b-open-source-ocr-revolution-2025"}, {"title": "Papers Explained 480: olmOCR 2 | by Ritvik Rastogi | Medium", "url": "https://ritvik19.medium.com/papers-explained-480-olmocr-2-33fbf89757eb"}, {"title": "olmOCR 2: Unit test rewards for document OCR | Ai2", "url": "https://allenai.org/blog/olmocr-2"}, {"title": "allenai/olmocr-2-7b-1025 ? LM Studio", "url": "https://lmstudio.ai/models/allenai/olmocr-2-7b-1025"}, {"title": "olmOCR-2-7B-1025 OCR Model", "url": "https://www.emergentmind.com/topics/olmocr-2-7b-1025"}, {"title": "allenai/olmocr-2-7b ? LM Studio", "url": "https://lmstudio.ai/models/allenai/olmocr-2-7b"}, {"title": "allenai/olmOCR-2-7B-1025 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025"}, {"title": "olmOCR 2", "url": "https://lmstudio.ai/models/olmocr2"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8"}]

Based on the search results, here's what I found about olmOCR 2 and its relationship to the Qwen2.5-VL-7B base model and reinforcement learning:

Model Architecture and Training

The olmOCR-2-7B-1025 VLM is initialized from a Qwen2.5-VL-7B-Instruct model fine-tuned on the olmOCR-mix-1025 dataset (a refreshed mix of 267,962 pages), and is further trained for one epoch on olmOCR2-synthmix-1025 using the Group Relative Policy Optimization (GRPO) algorithm via the Hugging Face TRL library.

The olmOCR-mix-1025 Dataset

The olmOCR-mix-1025 dataset contains 267,962 pages from over 100,000 PDFs sampled from diverse sources, including 9,828 pages from national archives. The new mix has been re-processed using GPT-4.1 instead of GPT-4o, has more consistent equation formatting (with $[$ and $($ for block and inline math), uses HTML format for tables, and includes basic alt text for images.

Reinforcement Learning with Verifiable Rewards

The reward signal for RL is binary-valued: for each document completion, a page-level reward is computed as the fraction of passing unit tests (from 0.0 to 1.0). Two additional binary rewards are incorporated: one for correct End-of-Sequence (EOS) token generation and another for the presence of document metadata (e.g., primary language, rotation correction factor) at the top of the response.

Model Performance

olmOCR 2 scores 82.4 points on the olmOCR-Bench, delivering substantial improvements where OCR often fails.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:28.222Z)

The primary source is confirmed. The claim's architectural assertions (Qwen2.5-VL-7B-Instruct base, fine-tuned then RL/GRPO trained) are well-supported. Now I need to verify the VRAM/hardware footprint sub-claim ? the "16-24 GB GPU class at FP16, less when quantized." Let me check the actual model card and quantization options.

7. user (2026-05-31T04:21:35.679Z)

Hardware Requirements for olmOCR-2-7B-1025

Based on the provided documentation, here's what I can determine:

Model Size

- **8B parameters** in BF16 (full precision)
- **Tensor type:** BF16

Quantized Versions Available

Yes, quantized versions exist:

- **FP8 version:** [olmOCR-2-7B-1025-FP8](https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8) is recommended for practical use
- **Other quantizations:** The documentation mentions "11 quantized models" supporting llama.cpp, LM Studio, Jan, and Ollama

Quote from documentation:

> "We recommend using the FP8 version for all practical purposes except further fine tuning."

GPU Requirements

The documentation does **not explicitly specify** GPU VRAM requirements or recommended GPU models.

However, you can infer rough requirements:

- **BF16 full model:** ~16GB VRAM (8B parameters × 2 bytes per parameter)
- **FP8 quantized:** ~8GB VRAM (approximately half the size)

Recommended Approach

For efficient production deployment, the documentation recommends:

> "The best way to use this model is via the [olmOCR toolkit](https://github.com/allenai/olmocr). The toolkit comes with an efficient inference setup via VLLM that can handle millions of documents at scale."

For detailed hardware specifications, you would need to consult the [olmOCR GitHub repository](https://github.com/allenai/olmocr) or the [paper](https://olmocr.allenai.org/papers/olmocr.pdf).

8. user (2026-05-31T04:21:37.521Z)

Web search results for query: "olmOCR-2-7B-1025 VRAM requirement GPU memory 16GB 24GB inference"

Links: [{"title": "Can You Run This LLM? VRAM Calculator (Nvidia GPU and Apple Silicon)", "url": "https://apxml.com/tools/vram-calculator"}, {"title": "Ollama VRAM Requirements: Complete 2026 Guide to GPU Memory for Local LLMs | LocalLLM.in", "url": "https://localllm.in/blog/ollama-vram-requirements-for-local-llms"}, {"title": "bartowski/allenai_olmOCR-2-7B-1025-GGUF · Hugging Face", "url": "https://huggingface.co/bartowski/allenai_olmOCR-2-7B-1025-GGUF"}, {"title": "FP8 model extremely slow on 12GB VRAM GPU due to FP16 upcast and hardcoded VRAM check · Issue #415 · allenai/olmocr", "url": "https://github.com/allenai/olmocr/issues/415"}, {"title": "GitHub - allenai/olmocr: Toolkit for linearizing PDFs for LLM datasets/training · GitHub", "url": "https://github.com/allenai/olmocr"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 - Fast, Reliable, and Scalable Inference on FriendlyAI", "url": "https://friendly.ai/model/allenai/olmOCR-2-7B-1025-FP8"}, {"title": "allenai/olmOCR-2-7B-1025 - Fast, Reliable, and Scalable Inference on FriendlyAI", "url": "https://friendly.ai/models/allenai/olmOCR-2-7B-1025"}, {"title": "Deploying olmOCR 2: High-throughput OCR on Spheron Decentralized GPU", "url": "https://blog.spheron.network/deploying-olmocr-2-high-throughput-ocr-on-spheron-decentralized-gpu"}, {"title": "GPU Memory Requirements for LLMs: VRAM Calculator | Spheron Blog", "url": "https://www.spheron.network/blog/gpu-memory-requirements-llm/"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Discussions", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8/discussions"}]

Based on the search results, here's what I found regarding VRAM requirements for the olmOCR-2-7B-1025 model:

VRAM Requirements

Consumer GPUs with 8 to 24 GB VRAM can run these models with suitable batch sizes. This indicates that the model can operate within the 16GB-24GB range you mentioned, though performance may vary depending on the specific variant and batch size.

Important Considerations

****FP8 Variant:**** The FP8 quantized model runs at approximately 0.5 tokens/second on a 12GB VRAM GPU, with memory usage increasing from ~7GB to ~20GB due to weights being upcast to FP16 during computation. This suggests that while FP8 weighs less, actual inference may require more memory.

****GGUF Quantized Versions:**** For the GGUF format, you should aim for a quantization with a file size 1-2GB smaller than your GPU's total VRAM to ensure optimal performance.

****Production Setup:**** Large-scale production setups use A100 or H100 GPUs with 40-80 GB VRAM, enabling higher batch throughput.

In summary, both 16GB and 24GB GPUs can run olmOCR-2-7B-1025, with 24GB providing more headroom for higher batch sizes and longer context lengths.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:21:44.009Z)

I now have strong corroboration. Let me confirm a couple of nuances: the model card lists "8B parameters" tensor count (Qwen2.5-VL-7B's actual param count is ~8.3B including the vision encoder, commonly branded "7B"), and the FP16/BF16 footprint. Let me verify the official olmOCR docs on the recommended minimum GPU to be thorough about the "16-24 GB" tier framing.

10. user (2026-05-31T04:21:47.633Z)

olmOCR GPU Requirements

Based on the documentation, here are the stated hardware requirements:

****Minimum Requirements:****

- "Recent NVIDIA GPU (tested on RTX 4090, L40S, A100, H100) with at least 12 GB of GPU RAM"
- 30GB of free disk space

****Tested GPUs:****

The system has been verified to work with RTX 4090, L40S, A100, and H100 models, indicating support for high-end consumer and enterprise-grade GPUs.

The documentation specifies a 12GB VRAM minimum but doesn't explicitly state "recommended" specs beyond the tested hardware examples. The 7B parameter vision language model underlying olmOCR appears designed for enterprise-scale GPU resources.

11. user (2026-05-31T04:21:48.539Z)

olmOCR 2 Base Model and Training

****Base Model:****

The blog post confirms olmOCR 2's foundation: "olmOCR 2 is built on Qwen2.5-VL-7B and fine-tuned on olmOCR-mix-1025, a dataset of 270,000 PDF pages with diverse properties."

Note: The post does not specify "Qwen2.5-VL-7B-Instruct" explicitly?it only mentions "Qwen2.5-VL-7B."

****Training Method:****

The blog describes reinforcement learning with GRPO: "We apply Group Relative Policy Optimization (GRPO), a reinforcement learning algorithm, using pass/fail rewards from these unit tests."

The training pipeline involved two stages: initial fine-tuning on olmOCR-mix-1025, followed by GRPO-based training using verifiable unit test rewards to optimize for specific correctness criteria (table structure, equation accuracy, reading order, etc.).

12. assistant (2026-05-31T04:22:01.285Z)

I now have comprehensive verification from multiple independent sources. Let me assess the claim against the checklist.

****Architectural claims (well-supported by primary source + corroborating sources):****

- "Built on Qwen2.5-VL-7B-Instruct, a 7B-parameter VLM" ? Confirmed by the arXiv paper, HuggingFace model card, Ai2 blog, and multiple secondary reviews. The exact quote matches.
- "fine-tuned then RL-trained" ? Confirmed. Paper: SFT on olmOCR-mix-1025, then GRPO (Group Relative Policy Optimization, a reinforcement learning algorithm).

****VRAM/hardware footprint sub-claim (the part requiring scrutiny):****

- "anchors its VRAM/hardware footprint to the 7B Qwen2.5-VL tier (roughly 16-24 GB GPU class at FP16, less when quantized)."
- The model card states 8B params in BF16 ? ~16 GB weights at FP16/BF16, which lands squarely in the "16-24 GB GPU class at FP16."
- FP8 version exists and is recommended ("less when quantized"), plus 11 quantized models (GGUF, Ollama, etc.) ? confirms "less when quantized."
- One important nuance: the official olmOCR toolkit states a ****12 GB minimum**** VRAM (tested on RTX 4090). This is slightly lower than the claim's "16-24 GB at FP16" framing ? but the 12 GB minimum is achievable via the FP8/quantized path (and GitHub issue #415 confirms the FP8 model is "extremely slow on 12GB" due to FP16 upcast). At true FP16/BF16 the ~16 GB weight footprint plus KV cache/activations genuinely needs the 16-24 GB tier. So the claim's qualifier "at FP16... less when quantized" is consistent with the 12 GB-via-quantization minimum.

The claim is accurate and appropriately hedged. The "7B" branding is the model's own naming (Qwen2.5-VL-7B-Instruct), even though the actual tensor count is ~8B including the vision encoder ? this is standard naming convention and the claim faithfully uses the model's own label. The VRAM tiering ("16-24 GB at FP16, less when quantized") is a reasonable, correctly-hedged inference directly supported by the BF16 model size and the existence of FP8/GGUF quantizations.

This claim survives refutation attempts. Sources are primary (arXiv paper, official model card, Ai2 blog) and current (October 2025 release, well within the research window). No contradicting evidence found.

13. user (2026-05-31T04:22:10.999Z)

Structured output provided successfully

14. assistant (2026-05-31T04:22:14.932Z)

Claim verified as ****not refuted**** with high confidence. The architectural assertions (Qwen2.5-VL-7B-Instruct base, fine-tuned then GRPO/RL-trained) are confirmed verbatim in the primary arXiv source and corroborated by the official Ai2 blog and HuggingFace model card. The VRAM tiering ("16-24 GB at FP16, less when quantized") is a correctly-hedged inference supported by the BF16 model size (~16 GB weights) and the existence of FP8/GGUF quantizations. The only nuance ? the toolkit's stated 12 GB minimum ? is reconciled by the claim's own "less when quantized" qualifier (the 12 GB path runs the quantized build).